Nutrition and colonic health: the critical role of the microbiota.

PURPOSE OF REVIEW: To highlight mechanisms whereby diet affects colonic function and disease patterns. RECENT FINDINGS: Topical nutrients are preferentially used by the gut mucosa to maintain structure and function. With the colon, topical nutrients are generated by the colonic microbiota to maintain mucosal health. Most importantly, short chain fatty acids control proliferation and differentiation, thereby reducing colon cancer risk. In patients with massive loss of small intestine, short chain fatty acid production supports survival by releasing up to 1000 kcal energy/day. Human studies show that the microbiota synthesizes a large pool of utilizable folate which may support survival in impoverished populations. Unfortunately, the microbiota may also elaborate toxic products from food residues such as genotoxic hydrogen sulfide by sulfur-reducing bacteria in response to a high-meat diet. The employment of culture-free techniques based on 16S regions of DNA has revealed that our colons harbor over 800 bacterial species and 7000 different strains. Evidence suggests that the diet directly influences the diversity of the microbiota, providing the link between diet, colonic disease, and colon cancer. The microbiota, however, can determine the efficiency of food absorption and risk of obesity. SUMMARY: Our investigations have focused on a small number of bacterial species: characterization of microbiota and its metabolism can be expected to provide the key to colonic health and disease.